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Behind the Words: Unmasking Bias Lurking in LLMs and Data

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***Abstract*- This study examines hidden biases in large language models (LLMs). Three popular models—RoBERTa, BERT, and GPT-2—were evaluated using StereoSet, CrowS-Pairs, and Indian-BhED, datasets designed to expose biases. Our comparative analysis reveals clear variations in how these models encode and reproduce bias, underscoring the difficulty of mitigation. The results highlight the need for bias-aware evaluation. By contributing empirical evidence, this work advances the discussion on ethical AI and points to directions for building more responsible language technologies.**

Keywords- Large Language Models (LLM), Bias, Bidirectional Encoder Representations from Transformers (BERT), Generative Pre-trained Transformer (GPT), RoBERTa

I. INTRODUCTION

Large Language Models (LLMs) have emerged as a transformative force in artificial intelligence, reshaping how people communicate, access information, and make decisions. Their ability to generate human-like text has enabled applications across education, healthcare, and public services. However, alongside these advantages comes a persistent concern—bias within their training data and outputs. Such biases, often subtle and systemic, can reinforce stereotypes or overlook marginalized groups, reflecting the inequalities embedded in human-generated data. If unaddressed, they risk producing unfair outcomes, misinformation, and reduced trust in AI systems.

This research investigates the presence of bias in three widely used LLMs—RoBERTa, BERT, and GPT-2—using three benchmark datasets: StereoSet, CrowS-Pairs, and Indian-BhED. These datasets assess model behavior across sensitive attributes such as gender, race, and socioeconomic background. Through comparative evaluation, the research highlights how biases manifest across models and datasets, contributing to a clearer understanding of the ethical challenges posed by large-scale language models.

The research pursues four objectives:

1. Identify and categorize biases present in LLMs.
2. Compare bias patterns across different models.
3. Assess dataset influence on the type and extent of bias.
4. Explore underlying mechanisms that give rise to such biases.

The findings underscore the complexity of mitigating bias in LLMs and emphasize the need for ongoing scrutiny and responsible deployment to ensure equitable AI outcomes.

In this study, we begin by reviewing and analyzing prior work conducted in this domain. We then outline the methodology employed in our research. Followed by analyzing the results that are generated and visualizing them for better understanding and conclusion.

II. LITERATURE REVIEW

While Large Language Models (LLMs) such as BERT, RoBERTa, and GPT-2 have achieved remarkable success across NLP tasks, they also encode social and cultural biases. Datasets like StereoSet, CrowS-Pairs, and Indian-BhED have been introduced to assess how these models handle stereotypical associations in diverse contexts.

A. Bias: Types, Sources, and Mitigation Challenges

According to Ranjan [1], bias in LLMs arises from three key sources—data, algorithms, and system interactions. Data bias reflects stereotypes present in training corpora, algorithmic bias stems from model design and optimization, and system bias results from user interactions and deployment contexts. Algorithms are products of human thought and social practice; they reflect the values, assumptions, and limitations of the communities that produce them[7]. Developers' cultural perspectives further influence fairness. Detecting and quantifying bias involves both qualitative feedback and quantitative metrics such as statistical parity, regression analysis, and benchmark datasets.

Resnik [5] notes that many biases are embedded in the very structure of LLMs, suggesting that genuine mitigation requires rethinking model design rather than incremental fixes. Joshi et al. [2] found that gender bias in Hindi text generation (87.8%) is far higher than in English (33.4%), highlighting the challenge of fairness in low-resource languages. Motoki [10] observed

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similar tendencies in ChatGPT, which exhibited measurable political bias toward left-leaning positions.

Roberto [8] emphasizes that understanding the sources of bias is more important than post hoc mitigation. Training data drawn from online sources, such as Wikipedia—where 87% of editors are male—can reinforce representational imbalances. Bias is also shaped by factors like domain diversity, corpus curation, temporal scope, and linguistic representation. The gap between high- and low-resource languages continues to widen, underscoring the need for multilingual inclusion to reduce dominance-related bias.

Lin [9] further argues that even large-scale web training cannot eliminate bias, as social and cultural patterns are inherently embedded in the data. Similarly, Gonen and Goldberg [12] demonstrate that common "debiasing" techniques in word embeddings often mask rather than remove gender bias. Even after adjustments, gendered words remain clustered in semantically feminine contexts (e.g., "nurse," "receptionist," "teacher"), indicating that deeper societal stereotypes persist within the embedding space. The work in [6] raises urgent caution against using publicly available LLMs in clinical settings as publicly available chatbot LLMs demonstrate significant biases, which may negatively impact patient outcomes without appropriate precautions.

Overall, existing studies reveal that biases in LLMs are structural, culturally dependent, and difficult to remove completely. Effective mitigation demands transparency in data collection, inclusive corpus design, and continued cross-linguistic evaluation.

B. Importance of Language and Culture in LLMs

The idea behind "gender-fair" language is that by minimizing disparities in how women and men are referred to in language, it could help reduce inequalities in their social status [3]. India has the world's second-largest population of internet users. AI authority is defined as the legitimized power of AI to influence human behavior without requiring sufficient evidence of the system's capabilities. In Shivani's [4] study, 79.2% of respondents expressed willingness to accept AI-driven decisions. This high level of acceptance persisted even in high-stakes situations, with 73% of participants indicating trust in AI for tasks such as medical diagnosis, loan approvals, and hiring decisions [4]. Traditional auditing methods, such as those conducted by researchers or activists, may fall short in identifying harmful behaviors when auditors lack the cultural background or lived experience needed to recognize certain issues as problematic [8]. Analysis performed in [14] shows that, without cultural prompting, GPT models align most closely with the cultural values of Anglosphere countries and Protestant Europe, while differing most from those of African-Islamic nations. Just as the association of "man" with "computer programmer" and "woman" with "homemaker" initially signaled gender bias in language models, the way responses from five popular GPT LLMs reflect the cultural values and self-expression of Western countries serves as evidence of cultural bias. Similarly work in [11] have highlighted that large language models (LLMs) often exhibit sociocultural and regional biases embedded within their training data

In summary, prior research has shown that large language models such as BERT, RoBERTa, and GPT-2 frequently inherit and reproduce social biases, and benchmark datasets like StereoSet, CrowS-Pairs, and Indian-BhED have proven effective for exposing these issues. However, most studies tend to evaluate models or datasets in isolation, limiting our understanding of how biases vary across both architectures and cultural contexts. This gap underscores the need for a systematic cross-model and cross-dataset comparison. To address these concerns, this research undertakes a detailed evaluation of three prominent large language models—RoBERTa, BERT, and GPT-2. The analysis draws on three benchmark datasets—StereoSet, CrowS-Pairs, and Indian-BhED—to provide a more comprehensive understanding of how bias appears and varies across these models. The next section describes the datasets in detail and explains the methodology adopted for this analysis.

III. Methodology: Framework for Evaluating Bias in LLMs

This work adopts a comparative framework to examine bias in large language models (LLMs), focusing on BERT, RoBERTa, and GPT-2, and evaluating them using three benchmark datasets: StereoSet, CrowS-Pairs, and Indian-BhED. Together, these datasets provide complementary perspectives on how biases manifest across different social dimensions. This is one of the initial researches where a comprehensive analysis of various LLMS as well as biases is compared especially using an Indian dataset to shed light on the biases generated due to lack of fair training models.

A. Models Considered

BERT is a bidirectional transformer commonly used for NLP tasks. RoBERTa is an optimized variant of BERT, trained on larger datasets with refined hyperparameters, often achieving higher performance. GPT-2 is an autoregressive model known for fluent text generation but prone to amplifying biases.

B. Dataset Used

StereoSet evaluates stereotypical biases in gender, profession, race, and religion. CrowS-Pairs contrasts stereotyping and non-stereotyping statements across nine categories, including race, religion, and age. Indian-BhED is designed for the Indian context, covering caste, religion, region, and intersectional biases, providing insight into localized societal patterns.

C. Bias patterns in LLMs

Bias in LLMs stems largely from training data and fine-tuning processes. Training corpora, drawn from books, websites, and other sources, reflect societal prejudices and cultural norms, which models can inadvertently learn. Human reviewers involved in fine-tuning can also introduce bias through subjective evaluations, influencing model outputs. Following are the bias patterns considered in this work:

Gender: Models often reinforce traditional roles, associating certain professions with specific genders and amplifying stereotypes.

Race and Ethnicity: Models may generate offensive language, culturally insensitive content, or misleading information.

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Socioeconomic and Educational: Behaviors and attitudes are sometimes linked to specific social classes, and perspectives from less formally educated groups are often underrepresented.

This framework enables systematic evaluation of above-mentioned bias in the current research work across models and social dimensions, providing a clearer understanding of how model architecture, training data, and fine-tuning practices influence bias in contemporary LLMs.

D. Evaluation Framework: Method for Bias Analysis

The research work follows three main stages: dataset preprocessing, model evaluation, and bias measurement.

1) Dataset Preprocessing

All datasets were standardized into formats compatible with the selected models. For masked language models (BERT and RoBERTa), inputs were tokenized using their respective tokenizers. For the autoregressive GPT-2, prompts were modified to align with masked prediction tasks. In the Indian-BhED dataset, question–answer pairs were reformatted as prompts with expected responses to ensure consistent evaluation across models.

2) Model Evaluation

Each model was tested on the prepared datasets following their respective evaluation protocols. In StereoSet, models completed sentences to compute the Language Modeling and Stereotype Scores. In CrowS-Pairs, log-probabilities for stereotypical and anti-stereotypical pairs were compared to record model preference. For Indian-BhED, predictions were analyzed against expected responses to identify cultural and stereotype-related bias.

3) Bias Measurement and Comparison

Dataset-specific metrics, including StereoSet (S-score, ICAT), CrowS-Pairs (accuracy and bias percentage), and Indian-BhED (QA response analysis), were calculated and compared across models. A custom function, analyze_bias, was implemented to measure model preference for stereotypical, anti-stereotypical, and unrelated responses. This function computes the likelihood of each candidate sentence using a token probability estimator, get_token_probability, which assigns scores to each candidate based on model predictions. The relative likelihoods of stereotypical versus anti-stereotypical responses were used to derive bias scores.

This method was applied to BERT, RoBERTa, and GPT-2 to capture architectural variations in bias expression. BERT and RoBERTa, with bidirectional transformers, provide contextual insights for masked tokens, while GPT-2, a unidirectional model, offers a generative perspective. Together, these analyses enabled a quantitative and comparative assessment of bias, forming a consistent framework for benchmarking bias in large language models.

IV. Results & Analysis

The evaluation of BERT, RoBERTa, and GPT-2 was conducted across four key dimensions of social bias: race, religion, gender, and caste. Each dimension represents a distinct aspect of societal inequality that may be reflected in model outputs.

Race: All three models exhibited measurable racial bias, GPT-2 with 62.26% showing approximately 35% stronger stereotypical associations than BERT and RoBERTa.

Religion: Bias related to religion was evident across all models. GPT-2 again demonstrated the highest bias 72.36%, followed by RoBERTa, while BERT showed comparatively lower—but still significant—levels of bias.

Gender: Gender bias was consistently observed, with the models often reinforcing traditional gender roles and stereotypes. GPT-2 and RoBERTa showed similar degrees of bias approximately 55%, whereas BERT exhibited 5% lower tendencies.

Caste: All three models revealed caste-related bias, though the intensity varied. RoBERTa showed the highest bias 61.32%, followed by GPT-2 44.82%, while BERT recorded the lowest. There was a difference of 40.57% between RoBERTa and BERT. These findings underscore the need for culturally specific benchmarks to better detect and mitigate such biases.

Overall, the results as shown in Table 1 indicate that while bias persists across all examined dimensions, its severity and manifestation vary depending on both the model architecture and the sociocultural factor under consideration. Table 2 showcases the bias score for stereoset and CrowS-Pairs dataset across all three large language models.

TABLE I. Comparison between different LLM models

File	RoBERTa	GPT2	BERT
Race	21.50	62.26	27.46
Religion	63.41	72.36	35.77
Gender	55.35	54.08	49.69
Caste	61.32	44.82	20.75

TABLE II. Bias Percentages for Stereoset and CrowS-Pairs datasets.

Dataset	RoBERTa	GPT2	BERT
stereoset dataset	51.71%	52.33%	44.49%
CrowS-Pairs	59.15%	57.63%	58.36%

Figure 1 shows the comparison of bias scores across the three datasets for all the large language models considered in the experimentation. GPT2 portrays higher bias with respect to RoBERTa and BERT.

Overall, this work adds to research on using AI responsibly and ethically. By comparing biases across models and datasets, we show that bias still exists in current systems and highlight the need for future work on better debiasing methods, culturally aware benchmarks, and transparent reporting. Such efforts are key to building language models that are both effective and fair in real-world use.

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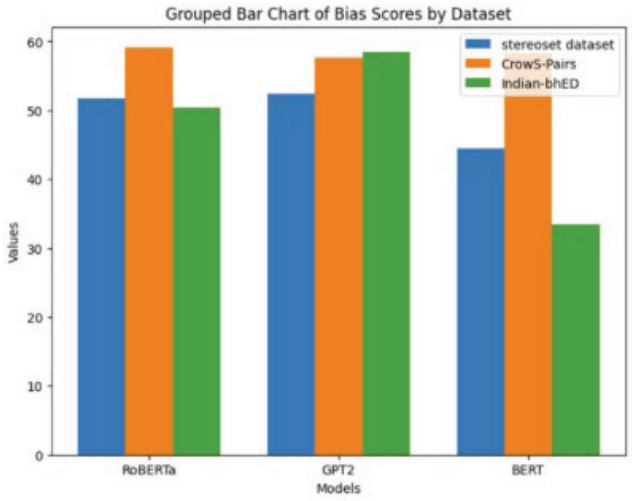


Fig. 1. Comparison of bias scores for RoBERTa, and GPT-2 across three datasets

V. Conclusion

This research work examined bias in three popular language models—BERT, RoBERTa, and GPT-2—using datasets like StereoSet, CrowS-Pairs, and Indian-BhED. The results showed that all models contain bias, but the points. type and level vary by model and dataset. StereoSet and CrowS-Pairs mainly revealed gender and racial stereotypes, while Indian-BhED highlighted cultural biases specific to India. These findings show that bias in LLMs is complex and shaped by different social and cultural contexts, making fair evaluation a difficult task.

The results highlight two main points. First, bias mitigation cannot follow a single universal approach, since cultural and linguistic contexts strongly affect how bias appears. Second, it is important to create and use culturally inclusive datasets to build fairer and more representative AI systems.

Overall, this work adds to research on using AI responsibly and ethically. By comparing biases across models and datasets, we show that bias still exists in current systems and highlight the need for future work on better debiasing methods, culturally aware benchmarks, and transparent reporting. Future work may involve examining a broader range of biases and integrating or comparing various debiasing techniques. Such efforts are key to building language models that are both effective and fair in real-world use.

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